

SMART BOREHOLE SAFETY SYSTEM

A MINI PROJECT REPORT

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ABSTRACT

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Unprotected boreholes are a serious safety hazard and can cause accidents to people and animals, especially in rural areas and construction sites. Traditional safety methods, like manual covers and occasional monitoring, are unreliable and cannot respond quickly during emergencies. To address this, a Smart Borehole Safety System is proposed, which provides continuous monitoring, automatic protection, and intelligent response. The system has two layers: an outer sensing layer and an inner protection layer. The outer layer uses four PIR sensors placed around the borehole to detect motion from all directions. Detected activity is processed by an ESP32 microcontroller, and instant alerts are sent through an IoT platform.

The inner layer controls a servo motor-driven safety cover that automatically opens or closes based on detected motion, eliminating the need for manual operation. A manual override with push buttons is included for emergencies and maintenance. Additional features include motion-triggered camera capture, a night-time safety mode, and a risk assessment system that classifies activity as low, medium, or high risk. The system is cost-effective, scalable, and practical for real-time use, providing an effective solution to prevent borehole-related accidents.

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CHAPTER 1

INTRODUCTION

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INTRODUCTION

Water is an essential resource for human survival, and boreholes play a crucial role in providing access to underground water, especially in rural and semi-urban areas, where they are widely used for drinking, agriculture, and daily needs, ensuring a steady and reliable water supply. With the rise in population and increasing water scarcity, the use of boreholes has grown rapidly. However, poor maintenance and lack of safety awareness have created serious public safety concerns, as many accidents occur due to unattended or improperly secured boreholes, posing risks to both humans and animals.

1.1 Background of the Study

Smart monitoring systems have become an important part of modern technology, designed to improve safety, efficiency, and reliability in various environments. These systems continuously observe surroundings using advanced components such as sensors and microcontrollers, allowing them to detect unusual activities or potential risks in real time. Unlike traditional methods that depend on manual supervision, smart monitoring systems provide automatic and accurate monitoring, reducing human effort and minimizing errors. This makes them highly useful in areas where safety is critical, such as construction sites, industrial zones, and remote locations. They also help in improving overall system performance and operational control.

One of the key advantages of smart monitoring systems is their ability to operate continuously without interruption. They provide 24/7 monitoring, ensuring that any changes or threats are detected immediately. Once a risk is identified, the system can send instant alerts or notifications, allowing quick response and preventive action. This real-time capability significantly reduces the

chances of accidents and enhances overall safety. In addition, these systems can store and analyse data, helping in identifying patterns and improving future decision-making processes. This ensures better planning and risk management in critical situations.

The integration of the Internet of Things (IoT) has further enhanced the effectiveness of smart monitoring systems. IoT connects physical devices through the internet, enabling them to collect, share, and process data efficiently. With the help of IoT, users can monitor and control systems remotely using smartphones or computers. This remote accessibility ensures that users can stay informed and take necessary actions from anywhere at any time. It also reduces the need for physical presence, saving time and resources while improving system management. As a result, it increases convenience and operational efficiency.

Another important feature of IoT-based smart monitoring systems is their flexibility and scalability. These systems can be easily expanded or modified based on the requirements of different applications. Whether it is a small-scale setup or a large industrial system, IoT technology allows seamless integration with other devices, mobile applications, and cloud platforms. This makes data access and management more convenient and efficient. Additionally, the cost-effectiveness of these systems makes them suitable for widespread adoption across different sectors. This adaptability makes them future-ready solutions.

Overall, smart monitoring systems combined with IoT technology play a vital role in creating safer and smarter environments. They enhance safety by providing real-time monitoring, quick alerts, and automated responses, while also improving efficiency through reduced manual effort and better data management. As technology continues to advance, these systems are expected to become even more intelligent and widely used in various fields. Thus, they form a strong foundation for the development of modern, secure, and automated systems that

contribute to improving the quality of life in society. They are becoming essential components of smart infrastructure development.

1.2 Problem Statement

Unprotected and unattended boreholes represent a significant safety hazard, particularly in rural areas and construction sites where proper safety measures are often neglected. These open boreholes can easily go unnoticed, especially in poorly lit or isolated environments, increasing the risk of accidental falls involving humans and animals. Many reported incidents highlight the seriousness of this issue, where victims, especially children, fall into deep boreholes, making rescue operations difficult and time-consuming. This not only threatens lives but also causes emotional and social distress in communities.

Existing safety measures for boreholes are mostly traditional and ineffective in preventing accidents. Common methods include covering boreholes with temporary lids or barriers and relying on periodic manual inspection. However, these approaches are unreliable as covers may be removed, damaged, or not properly secured. Manual monitoring also lacks consistency and fails to provide continuous observation, leaving boreholes exposed for long durations and increasing the chances of accidents.

Another major limitation of current systems is the absence of real-time monitoring and immediate response. In many cases, accidents are not detected quickly, leading to delays in rescue operations. Without automated detection systems, it becomes difficult to identify emergencies instantly. This delay can result in severe consequences, including serious injuries or loss of life, highlighting the need for a more responsive and intelligent safety solution.

Therefore, there is a strong need to develop an efficient and smart monitoring system that can address these issues effectively. The system should provide real-

time monitoring, quick detection of risks, and automatic safety actions to prevent accidents. By using technologies such as sensors, microcontrollers, and IoT-based communication, it can ensure continuous surveillance, instant alerts, and faster response during emergencies, thereby improving overall safety and reducing potential risks.

1.3 Objective

The primary objectives of this project are as follows:

- To enhance safety around open boreholes by preventing accidental falls of humans and animals through effective monitoring and protective measures. The system focuses on identifying potential risks at an early stage. It ensures that necessary safety actions are taken promptly. This helps in avoiding dangerous situations and improving overall safety in vulnerable areas.
- To implement real-time monitoring and alert systems using an IoT-enabled Flutter mobile application. The system allows users to receive instant notifications and continuous updates. This enables quick response and immediate action during risk situations. It also supports remote monitoring, making the system more efficient and user-friendly.
- To design and implement an automated protection mechanism that operates without human intervention. The system ensures continuous monitoring and safety around boreholes. It reduces dependency on manual supervision and minimizes human error. This improves reliability and ensures consistent protection in all conditions.
- To develop a cost-effective and scalable system that can be easily deployed in rural and urban areas. The design ensures affordability and adaptability for different environments. It allows easy integration with additional

sensors and technologies if required. This makes the system suitable for large-scale implementation and long-term usage.

- To improve emergency response efficiency by enabling faster detection and immediate alert generation. The system helps reduce response time during critical situations. It supports timely rescue actions and minimizes the severity of accidents. This contributes to saving lives and enhancing overall safety management.

1.4 Scope of the Project

The scope of this project focuses on designing and developing a smart borehole monitoring and safety system using IoT technology. The system aims to provide continuous real-time monitoring by using sensors and microcontrollers to detect motion or unusual activities near open boreholes. It is designed to automatically identify potential risks and take immediate action to prevent accidents. This helps in improving safety, especially in rural areas and construction sites where such hazards are common.

Another important aspect of the project scope is the implementation of a real-time alert system integrated with a mobile application. This allows users to receive instant notifications and monitor the status of the borehole remotely from anywhere. The system reduces the need for manual supervision and ensures quick response during emergency situations. By enabling remote access and automated alerts, the project enhances efficiency and reliability in managing safety.

In addition, the project includes the development of a cost-effective and scalable solution that can be easily deployed in various environments. It allows future enhancements such as integration with cloud platforms, additional sensors, and advanced technologies like AI for predictive analysis. The system is designed to be flexible, user-friendly, and adaptable, making it suitable for large-scale

implementation. Overall, the project aims to provide a practical and efficient solution for improving safety and preventing accidents related to open boreholes.

1.5 Social Impact

The proposed system has a significant positive social impact by improving safety around hazardous areas and reducing the risk of accidents involving humans and animals. It helps protect lives, especially in rural and semi-urban regions where such dangers are more common and safety measures are often limited. By providing continuous monitoring and timely alerts, it increases awareness among people and ensures a quick response during emergency situations, thereby minimizing the chances of serious injuries or loss of life. Additionally, it reduces dependence on manual supervision and encourages the adoption of modern technology for public safety. The system also contributes to building safer communities, supports responsible infrastructure management, and enhances the overall quality of life by creating a more secure and protected environment for society.

1.6 Challenges

The system faces several challenges that need to be carefully managed for effective operation. One of the main challenges is ensuring accurate motion detection without false alarms caused by animals or environmental factors such as heat and wind. Reliable internet connectivity is essential for real-time alerts, but it may be difficult to maintain in rural areas. Providing a continuous power supply for uninterrupted functioning is another important concern. Environmental conditions like rain, dust, and temperature changes can affect the performance of sensors and other components. Proper installation and calibration of sensors are required to achieve complete coverage and accurate detection. In addition, efficient integration of hardware and software, quick response during

emergencies, and maintaining long-term reliability of the system are also significant challenges.

1.7 Organization of the Report

The remainder of this report is organized as follows.

Chapter 1 introduces the background of the study, clearly defining the problem statement along with the objectives and scope of the research.

Chapter 2 provides a comprehensive literature survey of existing disaster detection methods, covering classical techniques, deep learning models, and hybrid approaches, while also identifying key research gaps.

Chapter 3 presents the proposed methodology in detail, including the overall system architecture, data preprocessing steps, individual model components, and the fusion strategy employed to integrate them.

Chapter 4 focuses on experimental results, offering performance evaluation and a comparative analysis to demonstrate the effectiveness of the proposed system.

Chapter 5 concludes the study by summarizing the major contributions and suggesting potential directions for future research.

The report is completed with references and a detailed mapping of the project outcomes to Program Outcomes (PO), Program Specific Outcomes (PSO), and Sustainable Development Goals (SDGs).

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

The rapid advancement of Internet of Things (IoT) technology has significantly contributed to the development of intelligent monitoring and safety systems. These systems are widely used in real-time applications to improve safety, reduce human intervention, and provide immediate responses during critical situations. IoT-based systems integrate sensors, microcontrollers, and communication networks to continuously monitor environments and detect potential risks. Researchers have extensively explored sensor-based and microcontroller-based solutions to enhance safety in various domains such as industrial environments, construction sites, public infrastructure, and smart home systems [7], [13].

One of the most commonly used technologies in safety monitoring systems is the Passive Infrared (PIR) sensor. PIR sensors detect motion by sensing infrared radiation emitted by humans and animals, making them highly effective for surveillance and security applications. These sensors are widely preferred due to their low power consumption, cost-effectiveness, and high sensitivity. Jatiyoso et al. [1] and Aprilia & Purwanto [10] developed PIR sensor-based systems integrated with ESP32-CAM modules for IoT-based monitoring. Their systems demonstrated efficient motion detection and real-time monitoring capabilities, highlighting the importance of combining sensing and communication technologies for enhanced safety applications.

The ESP32 microcontroller plays a crucial role in IoT-based monitoring systems due to its built-in Wi-Fi capability, high processing speed, and affordability. It enables seamless data transmission and real-time communication between devices and users. Premalatha [2] and OkoKpujie & Okokpujie [9]

designed ESP32-based monitoring systems that provide real-time alerts during hazardous situations. These studies demonstrated how microcontrollers can effectively process sensor data and trigger alerts instantly, improving response time and minimizing risks. The ability of ESP32 to support multiple sensors and communication protocols makes it a preferred choice for safety-critical applications.

In addition, multi-sensor IoT systems have gained attention for their ability to provide comprehensive monitoring and early warning. Ariyanto et al. [3] implemented an ESP32-based multi-sensor system for environmental monitoring, which demonstrated effective real-time data acquisition and analysis. Similarly, Bagaskara et al. [12] proposed a system integrating PIR sensors and ESP32-CAM for enhanced security monitoring. These works highlight the importance of combining multiple sensing technologies to improve detection accuracy and system reliability in complex environments.

Real-time alerting is another essential feature of IoT-based safety systems. Immediate communication between the system and users is critical for reducing response time during emergencies. Tahemeen and Patil [4] proposed an IoT-based accident detection system that sends instant notifications to users when a critical event occurs. Upagade et al. [13] also developed an IoT-based home security system with alert mechanisms. These systems emphasize the importance of fast and reliable communication in ensuring timely action and preventing further damage.

Researchers have also explored multi-layer safety mechanisms to improve system reliability and reduce false alarms. Supriyanto et al. [5] developed a smart monitoring system for construction workers using IoT technology, while Hasib et al. [16] proposed a dual-modality IoT framework for integrated safety monitoring. These systems integrate multiple safety layers and communication

modules to provide continuous monitoring and enhanced protection, making them suitable for hazardous environments.

Automation is a key aspect of modern safety systems, as it allows immediate action without human intervention. Automated systems significantly reduce response time and improve safety outcomes. Ullah et al. [6] designed an IoT-based early warning system that detects dangerous situations and automatically generates alerts. Similarly, Ajmal et al. [14] developed an anti-theft flooring system using ESP32 with automated response mechanisms. These systems demonstrate how integrating sensors with automation improves system efficiency and reliability.

The integration of IoT in security and surveillance systems has further improved monitoring capabilities. Traditional systems were limited to local monitoring, whereas modern IoT-based systems provide remote access and control. Jatiyoso et al. [1] and Fikri [11] demonstrated that combining motion detection with IoT connectivity significantly enhances security efficiency. This integration allows users to monitor environments in real time from remote locations, improving overall system effectiveness.

IoT technology is also widely used in smart infrastructure and home automation systems. Syari et al. [7] discussed the application of IoT in smart systems to improve security and convenience. Their study showed that IoT-based systems can continuously monitor environments and provide real-time updates, making them suitable for safety-critical applications. These systems enhance user convenience while ensuring safety through automation and remote access.

In industrial applications, IoT-based systems are increasingly used to ensure operational safety and efficiency. Tumin et al. [8] developed an ESP32-based automated system with enhanced safety features for industrial use. Their work highlighted the importance of integrating automation and IoT technologies

to reduce risks and improve control in real-world environments. Industrial safety systems benefit greatly from real-time monitoring and automated responses.

From the reviewed literature, it is evident that most existing systems focus on motion detection, real-time monitoring, wireless communication, and automated alert mechanisms. The combination of PIR sensors, ESP32 microcontrollers, and IoT platforms has proven to be highly effective in designing smart safety systems. These technologies enable continuous monitoring, instant alerts, and remote access, which are essential for preventing accidents in hazardous environments.

However, despite these advancements, several challenges remain in existing systems. Many IoT-based systems depend heavily on stable internet connectivity, which may not be reliable in remote or rural areas. False detections caused by environmental factors and animal movement are also common issues. Additionally, most systems require proper calibration and regular maintenance to ensure long-term performance.

Another important observation is that while many systems focus on detection and alerting, fewer systems provide integrated solutions that combine monitoring, automated protection, and layered safety mechanisms. This indicates a research gap in developing systems that not only detect risks but also take immediate preventive actions.

To address these limitations, there is a need for a comprehensive safety system that integrates real-time monitoring, automated response, and reliable communication. The proposed system focuses on combining PIR sensors, ESP32 microcontroller, and IoT technology with an automated protection mechanism to ensure effective accident prevention.

Furthermore, the proposed system aims to enhance accuracy and reliability by reducing false detections and ensuring consistent performance. Advanced techniques such as optimized sensor placement and multi-layer safety logic can improve system efficiency. The integration of a mobile application allows users to monitor the system remotely and receive instant alerts.

Overall, the literature survey highlights the significant role of IoT-based monitoring systems in improving safety and automation. While existing systems provide effective detection and alerting mechanisms, there is a need for more integrated and automated solutions. The proposed project aims to bridge this gap by developing a smart borehole safety system that combines monitoring, alerting, and automated protection.

In addition to the above aspects, future developments in IoT-based safety systems are expected to focus on improving intelligence and adaptability. The integration of artificial intelligence and machine learning can help predict potential risks based on historical data. Advanced communication technologies, energy-efficient designs, and user-friendly mobile applications will further enhance system performance and usability, making these systems more reliable and suitable for real-world applications.

The summary of the reviewed literature, including the methods used by various authors and their limitations, is presented in Table 2.1

Table 2.1: Summary of Selected Related Works

Authors	Method	Limitation
Jatioso et al. (2025) [1]	PIR sensor with ESP32 CAM for IoT-based monitoring	Limited to motion detection; may produce false alarms due to environmental factors
Premalatha (2024) [2]	Attention-based U-Net for landslide extraction	Requires high-quality labeled data; computationally intensive
Ariyanto et al. (2023) [3]	Fully Convolutional Neural Network (FCNN) for landslide detection	Limited ability to capture fine spatial details
Tahemeen & Patil (2024) [4]	LiDAR + Attention U-Net hybrid model	Requires LiDAR data, which is not always available
Supriyanto et al. (2024) [5]	SE-YOLOv7 with attention mechanisms	Improved accuracy but increased model complexity
Ullah et al. (2024) [6]	ShapeFormer (Transformer-based model)	High computational requirements; slower inference
Syari et al. (2024) [7]	Deep learning with object-based image analysis.	High computational cost; complex preprocessing required

Tumin et al. (2025) [8]	AI-based landslide prediction using satellite imagery.	Focuses on prediction; lacks real-time detection capability.
Okokpujie, K., & Okokpujie, I. P. [9]	Development of an affordable real-time IoT-based surveillance system using ESP32 and TWILIO API	Depends on internet connectivity; may face delays in notification delivery.
Aprilia, M. W., & Purwanto, A. [10]	IoT-based boarding room security system using PIR sensor and ESP32-CAM with Telegram notification	Limited to indoor security; performance affected by lighting and network issues
Fikri, A. [11]	Simple IoT-based home security system using ESP32 and Blynk	Basic system with limited scalability and fewer advanced safety features
Bagaskara, et al. [12]	Design of IoT-based security system using ESP32-CAM and PIR motion sensor in home environment	Prone to false detection due to environmental conditions and motion noise
Upagade, M., Ajapuje, A., Dongre, J., Kamble, R., & Panse, U. [13]	IoT based home security system.	Limited range and lacks advanced automation or multi-layer safety mechanisms
Ajmal, K., Muthiyil, A. S., Haneefa, J., Nihal, M., & Reshmi, C. S. [14]	Anti-theft flooring system using ESP32.	Application-specific; not suitable for broader environmental monitoring
Hasib, A., Akib, A. S. M., Ankur, N. D., & Giri, A. [15]	Dual-modality IoT framework for integrated access control and environmental safety monitoring	Complex system design; higher cost and implementation difficulty.

2.1 Findings

The review of existing literature indicates that IoT-based safety systems have significantly evolved with the integration of sensors, microcontrollers, and wireless communication technologies. Most of the studies focus on motion detection, real-time monitoring, and alert mechanisms using components such as PIR sensors and ESP32 microcontrollers. These systems are effective in providing immediate notifications and enabling remote monitoring, which enhances safety and reduces response time. Additionally, the use of IoT platforms allows continuous data transmission and visualization, making the systems more efficient and user-friendly. However, many existing solutions are limited to basic detection and alerting, without incorporating advanced automation or multi-layer safety features.

Furthermore, the analysis highlights several limitations and research gaps in current systems. A major issue is the dependency on stable internet connectivity, which can affect system performance in remote or rural areas. False alarms due to environmental conditions and lack of intelligent filtering mechanisms also reduce system reliability. While some studies explore advanced techniques such as artificial intelligence and multi-sensor integration, they often involve high computational complexity and increased cost. Therefore, there is a need for a more reliable, cost-effective, and integrated system that combines real-time monitoring, automated response, and improved accuracy. This motivates the development of the proposed smart borehole safety system, which aims to address these challenges effectively.

CHAPTER 3

METHODOLOGY

CHAPTER 3

METHODOLOGY

This chapter presents the methodological framework adopted for the design and implementation of the proposed smart borehole monitoring and safety system. It explains the overall system architecture, hardware and software components, data acquisition methods, and the integration of IoT technologies to achieve effective real-time monitoring and automated safety. The proposed system combines sensor-based detection, microcontroller processing, and wireless communication to ensure continuous observation of borehole environments. Each component of the system is carefully designed to enhance detection accuracy, reduce response time, and provide reliable safety measures. The methodology focuses on overcoming the limitations of traditional monitoring systems by introducing automation, remote accessibility, and instant alert mechanisms, thereby ensuring improved safety and efficiency in real-world conditions.

Building upon this framework, the system follows a structured workflow for efficient monitoring and control. The input is collected through sensors such as PIR sensors, which detect motion near the borehole. The sensed data is processed by the ESP32 microcontroller, which acts as the central control unit of the system. Based on the sensor input, the system identifies potential risks and triggers appropriate actions, such as activating alerts or safety mechanisms. The processed data is transmitted through IoT communication to a mobile application developed using Flutter, enabling users to receive real-time notifications and monitor the system remotely. Additionally, an automated protection mechanism is integrated to respond instantly without human intervention. Finally, the system outputs are displayed through a user-friendly interface, ensuring clear understanding and effective decision-making, thereby enhancing overall safety and reliability.

3.1 System Architecture



Fig 3.1: System Architecture of the Proposed Smart Borehole Safety System

The proposed system is designed as an IoT-based smart borehole monitoring and safety system that integrates sensor technology, microcontroller processing, and automated protection mechanisms to ensure real-time safety and accident prevention. The architecture follows a layered design approach consisting of sensing, processing, communication, and response layers, where each component works collaboratively to enhance system reliability and performance, as illustrated in Fig. 3.1.

The workflow begins with input from PIR sensors and monitoring units placed around the borehole to detect motion or unusual activity. The detected signals are transmitted to the ESP32 microcontroller, which acts as the central processing unit of the system. Based on the sensor data, the system identifies potential risks and triggers appropriate actions such as activating a buzzer, LED

indications, and an automated closing mechanism using a servo motor to prevent accidents. Simultaneously, the processed data is sent through IoT communication to a mobile application, where users receive real-time alerts and status updates such as safe condition, warning alerts, or critical alerts.

This system ensures continuous monitoring through an inner and outer layer safety mechanism, where the inner layer provides automated physical protection and the outer layer handles alert notifications and user interaction. The integration of real-time detection, automated response, and remote monitoring ensures quick action during emergencies. This layered and integrated architecture improves system efficiency, reduces human intervention, and provides a reliable and scalable solution for enhancing safety around open boreholes.

3.2 Methodology

The methodology of the proposed system is based on an IoT-enabled real-time monitoring and automated safety framework designed to prevent accidents near open boreholes. The system integrates sensors, a microcontroller, communication modules, and an automated protection mechanism to ensure continuous monitoring and immediate response. The PIR and ultrasonic sensors detect motion and object proximity, and the collected data is processed by the ESP32 microcontroller to identify potential risk conditions. Based on the analysis, the system activates safety measures such as buzzer alerts, LED indications, and automatic closure of the borehole cover using a servo motor. Simultaneously, the system transmits real-time data to an IoT platform, enabling remote monitoring through mobile or web applications. This methodology ensures quick detection, minimal response time, and reliable accident prevention.

3.2.1 Data Acquisition

The system is designed to collect real-time environmental data using a combination of PIR (Passive Infrared) and ultrasonic sensors strategically

placed around the borehole area. The PIR sensor works by detecting variations in infrared radiation emitted by living beings such as humans and animals, allowing it to identify motion within its range. This helps in recognizing any movement near the borehole, which could indicate potential danger.

At the same time, the ultrasonic sensor measures the distance of nearby objects by emitting ultrasonic waves and calculating the time taken for the echo to return. This enables accurate detection of objects or obstacles approaching the borehole. Together, these sensors continuously monitor the surroundings and generate input signals based on motion and distance changes. The specifications and functions of these sensors are summarized in Table 3.2.

Table 3.2: Sensor Specifications (for Data Acquisition)

Sensor	Type	Function	Range	Output
PIR Sensor	Motion Sensor	Detects human/animal movement	~5–7 meters	Digital (HIGH/LOW)
Ultrasonic Sensor	Distance Sensor	Measures object distance	2 cm – 400 cm	Distance in cm

3.2.2 Data Processing

The ESP32 microcontroller plays a crucial role in processing the input signals received from the PIR and ultrasonic sensors by continuously analysing motion detection and distance measurements in real time. It applies predefined threshold values and logical conditions to interpret the sensor data and classify the surrounding environment into either a SAFE state, where no significant movement or nearby objects are detected, or an ALERT state, where motion is identified or an object is detected within a critical distance range. The processing unit executes embedded control algorithms to filter noise, validate sensor

readings, and ensure accurate decision-making. Based on this analysis, the ESP32 triggers appropriate responses such as activating alarms, sending notifications, or initiating safety mechanisms, thereby ensuring timely and reliable system operation.

3.2.3 Decision Making

Based on the processed data obtained from the sensors, the system performs intelligent decision-making to determine appropriate safety actions in real time. The ESP32 microcontroller evaluates both motion detection from the PIR sensor and distance measurements from the ultrasonic sensor against predefined threshold values. If any movement is detected near the borehole or if an object is identified within a critical proximity range, the system immediately switches to ALERT mode, indicating a potential risk. In this state, necessary actions such as triggering alarms, activating warning indicators, or sending notifications are initiated to ensure safety. Conversely, if no motion or nearby objects are detected, the system remains in SAFE mode, indicating normal conditions. This structured decision-making process helps in accurately identifying hazardous situations while minimizing false alarms caused by environmental noise or minor disturbances, thereby improving the overall reliability and efficiency of the system. The different system conditions, corresponding sensor inputs, states, and actions are summarized in Table 3.3.

Additionally, the system continuously updates its status by re-evaluating sensor inputs at regular intervals to ensure real-time responsiveness. It can also incorporate delay mechanisms or multiple readings to confirm a threat before switching to ALERT mode, further reducing false triggers. This adaptive decision-making enhances system stability and ensures reliable performance under varying environmental conditions.

Table 3.3: System States (for Decision Making)

Condition	Sensor Input	System State	Action
No motion	No detection	SAFE	System continues monitoring
Motion detected	PIR triggered	ALERT	Buzzer and LED ON
Object near borehole	Ultrasonic detects close object	CRITICAL	Servo closes cover

3.2.4 Actuation Mechanism

The actuation module is responsible for executing physical actions based on the decisions made by the processing unit, thereby ensuring immediate response to potential hazards. Once the ESP32 determines the system state, it sends control signals to various output components to perform appropriate actions. LED indicators are activated to provide clear visual feedback, where different colors (such as green for SAFE and red for ALERT) indicate the current system status. Simultaneously, a buzzer is triggered to generate an audible warning, alerting nearby individuals about potential danger. In critical situations, the servo motor is automatically operated to close the borehole cover, providing a mechanical safety barrier and preventing accidents. This integrated actuation mechanism eliminates the need for human intervention and ensures rapid, reliable, and effective safety measures. The functions and responses of the actuation components are summarized in Table 3.4.

Table 3.4: Actuation Components (for Actuation Module)

Component	Function	Response
LED (Green/Red)	Visual indication	Shows SAFE / ALERT
Buzzer	Audible alert	Produces warning sound
Servo Motor	Mechanical action	Closes borehole cover

3.2.5 Communication and IoT Integration

The system incorporates Wi-Fi connectivity through the ESP32 microcontroller to enable seamless communication with an IoT platform such as ThingSpeak for real-time data transmission and monitoring. Sensor data, including motion detection and distance measurements, along with the current system status (SAFE, ALERT, or CRITICAL), are continuously sent to the cloud platform. This allows users to remotely monitor the borehole environment from anywhere using a web interface or mobile device. In addition to real-time monitoring, the platform provides instant alerts whenever a potential risk is detected, ensuring timely awareness and response. Furthermore, ThingSpeak supports data visualization features such as graphs and logs, which help in analysing trends, understanding system behaviour over time, and improving decision-making. The communication parameters and functionalities are summarized in Table 3.5.

Additionally, the system ensures secure and reliable data transmission using standard communication protocols supported by the ESP32. It also enables continuous data logging, which can be useful for future analysis and system optimization.

Table 3.5: Communication Details (for IoT Module)

Parameter	Description
Platform	ThingSpeak
Communication Type	Wi-Fi
Data Sent	Sensor values, system status
Output	Graphs, alerts, logs

3.2.6 System workflow

The system follows a continuous monitoring loop where sensors detect environmental changes, the microcontroller processes the data, decisions are made based on predefined logic, and appropriate actions are executed. The system also ensures real-time communication with the IoT platform, providing a complete and efficient safety monitoring solution.

CHAPTER 4

RESULTS AND DISCUSSION

CHAPTER 4

RESULT AND DISCUSSION

This system leverages IoT technology to provide a smart and efficient borehole safety solution by integrating real-time monitoring, automated alerts, and protective mechanisms. Using sensors such as PIR and ultrasonic, the system continuously observes the surroundings to detect human or animal movement and measure proximity to the borehole. Once a potential risk is identified, the system responds instantly by activating alarms, visual indicators, and safety mechanisms like closing the borehole cover, thereby preventing accidents. Through Wi-Fi connectivity, real-time data is transmitted to an IoT platform, enabling users to monitor conditions remotely and receive immediate alerts. This combination of intelligent sensing, fast response, and remote accessibility significantly enhances safety, reduces manual intervention, and ensures timely preventive action in hazardous situations.

4.1 Experimental Setup

The experimental evaluation of the proposed system is designed to assess the performance, reliability, and responsiveness of the IoT-based smart borehole monitoring system under real-time conditions. The study is structured in two phases: Phase 1 evaluates individual modules independently, including the sensing module (PIR and ultrasonic sensors for motion and distance detection), processing module (ESP32 for decision-making), and actuation module (buzzer, LED, and servo motor for safety response). Phase 2 evaluates the complete integrated system, where all modules work together with IoT communication to ensure real-time monitoring, automated protection, and remote alerting. Additionally, various test scenarios are conducted under different environmental conditions to validate system accuracy and consistency. The results obtained are

analyzed to measure system efficiency, response time, and overall effectiveness in preventing potential accidents.

The experiments were conducted using the following hardware configuration:

- ESP32 microcontroller
- PIR sensor for motion detection
- Ultrasonic sensor for distance measurement
- Servo motor for automatic borehole cover operation
- Buzzer and LED indicators for alert generation
- Power supply unit (5V/3.3V)

The system was implemented using the following software environment:

- Arduino IDE for programming the ESP32
- Embedded C/C++ for control logic implementation
- ThingSpeak IoT platform for cloud data storage and visualization
- Wi-Fi communication for real-time data transmission
- Flutter-based mobile application for remote monitoring (optional)

The system was tested under various real-time scenarios, including no motion (safe condition), human or animal movement near the borehole (alert condition), and close proximity detection (critical condition). Each scenario was evaluated based on system response time, accuracy of detection, and effectiveness of safety actions. The system continuously monitored sensor inputs, processed data, and triggered appropriate responses such as activating alerts and closing the borehole cover automatically.

Performance evaluation was carried out using metrics such as detection accuracy, response time, system reliability, and alert efficiency. Multiple test runs were conducted under different environmental conditions to ensure consistent performance. The results were analyzed based on the system's ability to detect

risks, generate timely alerts, and execute automatic protection mechanisms without delay.

This experimental setup ensures that the proposed system is thoroughly evaluated for real-world applicability, demonstrating its effectiveness in preventing accidents and providing a reliable, efficient, and scalable safety solution for borehole monitoring.

4.2 Results

The output results of the proposed IoT-based smart borehole safety system demonstrate its effective real-time monitoring, detection, and alert mechanisms. The system outputs are observed through multiple interfaces including the serial monitor, ThingSpeak cloud platform, and mobile notifications, as illustrated in Fig. 4.1, Fig. 4.2, and Fig. 4.3.

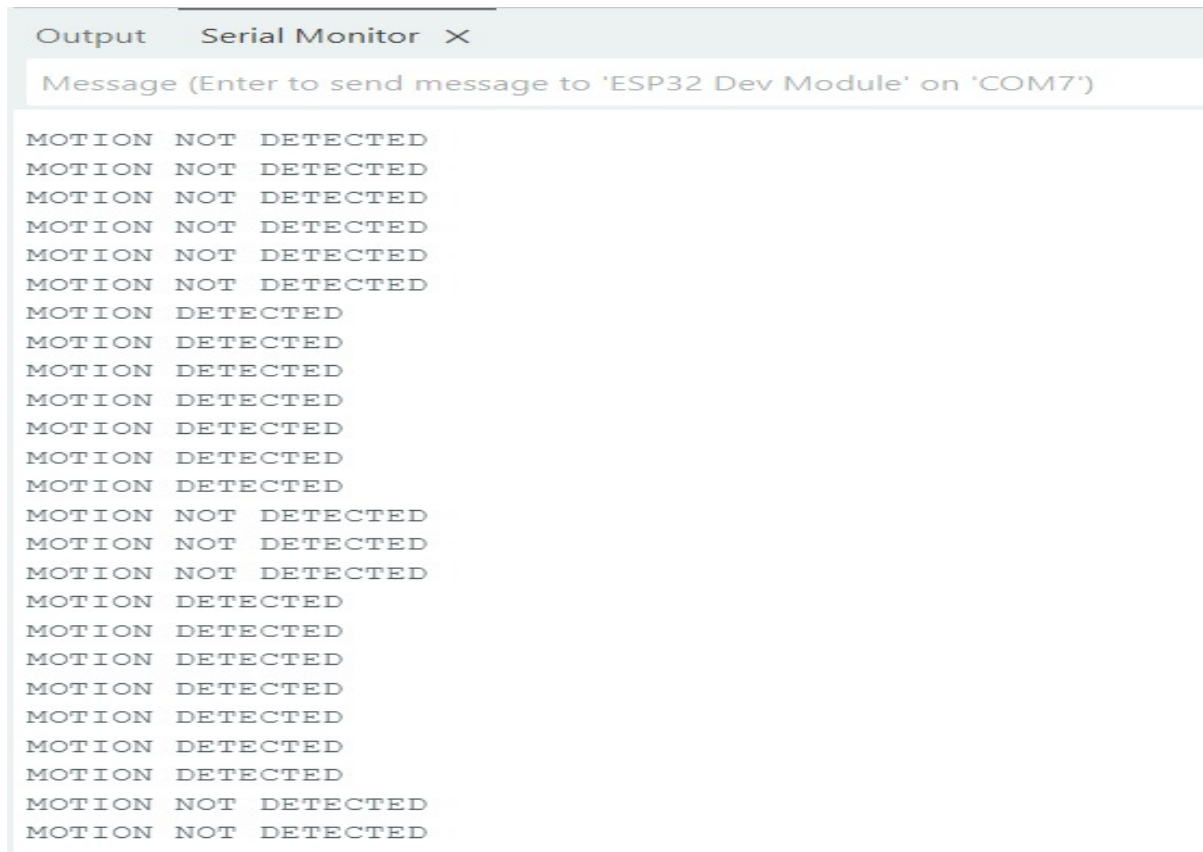
Fig. 4.1 shows the serial monitor output, where real-time system status such as motion detection and gate position (open/closed) is continuously displayed. This helps in verifying the internal working of the system and monitoring sensor responses during operation.

Fig. 4.2 represents the ThingSpeak output, where sensor data such as motion status and gate condition are visualized in the form of graphs. This enables users to analyze system behavior over time and ensures remote monitoring through the cloud platform.

Fig. 4.3 illustrates the mobile notification output, where instant alerts are sent to the user via Telegram when motion is detected. This ensures immediate awareness and allows quick action to prevent accidents.

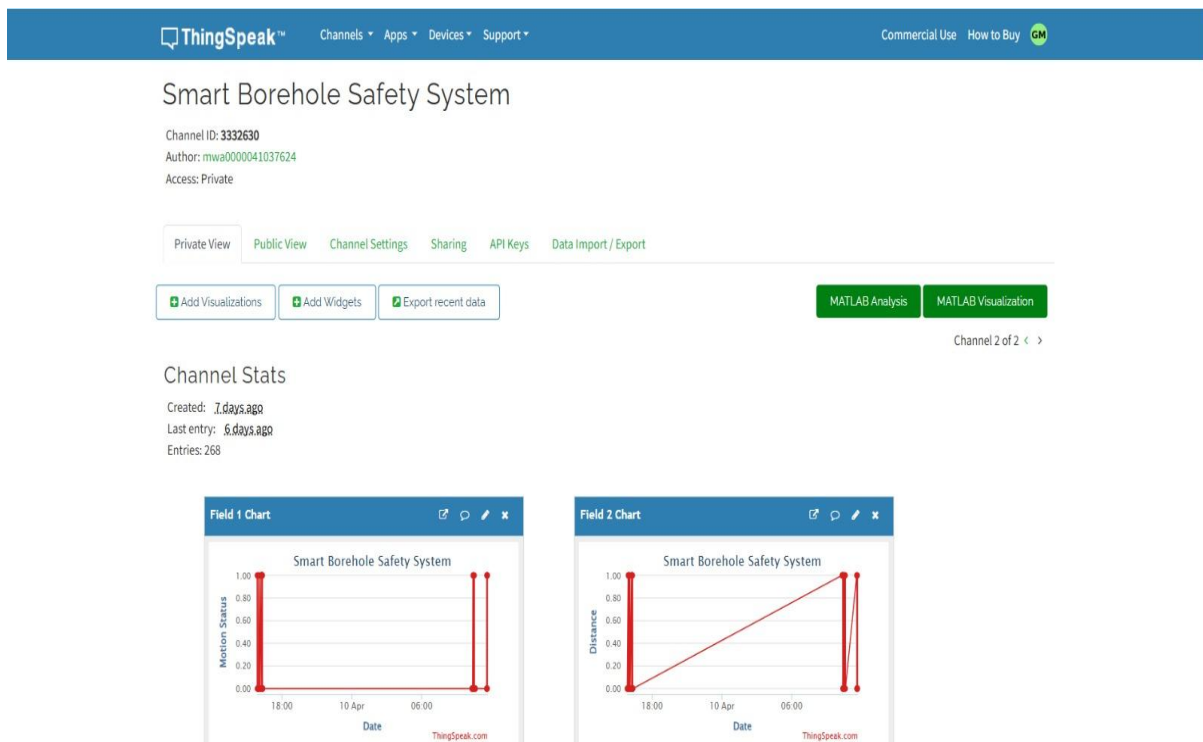
These outputs collectively confirm that the system successfully performs real-time detection, automated response, and remote monitoring, making it reliable and efficient for borehole safety applications.

Fig 4.1 :Serial Monitor Output



The serial monitor output displays continuous real-time updates from the system, indicating the status of motion detection. Messages such as “MOTION DETECTED” and “MOTION NOT DETECTED” are printed repeatedly, reflecting the input received from the motion sensor. This live data stream helps in validating whether the sensor is accurately detecting movement and responding to environmental changes. By observing these outputs, users can easily monitor system behavior, debug potential issues, and ensure that the motion detection mechanism is functioning reliably during operation.

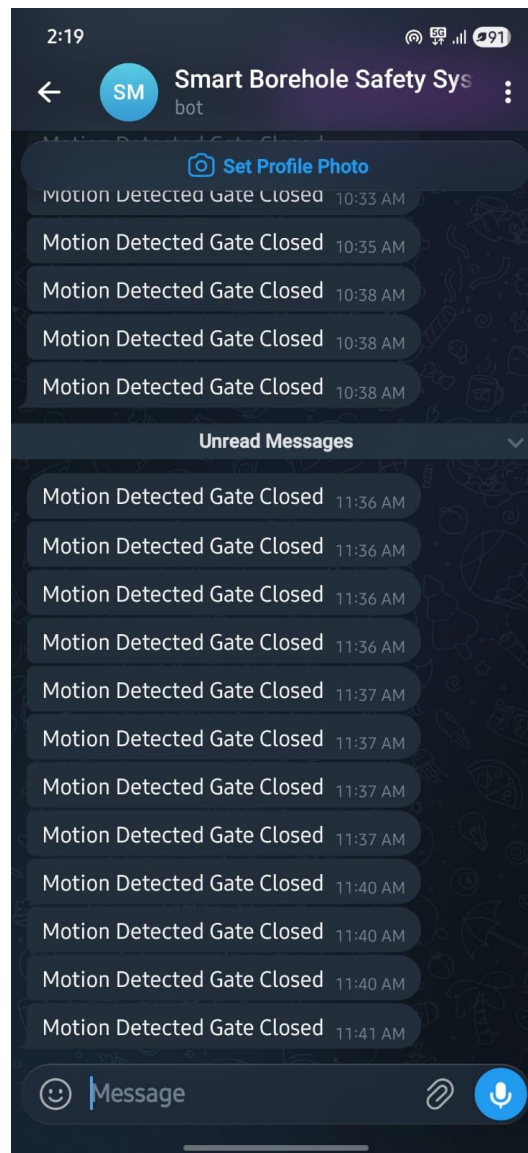
Fig 4.2 : ThingSpeak Output



The ThingSpeak output displays real-time data updates from the system in the form of graphs, representing parameters such as motion status and gate condition. The plotted values continuously change based on sensor inputs, showing when motion is detected and how the gate responds over time. This visual representation helps in verifying that the system is correctly sending data to the cloud and that sensor readings are accurately recorded. By observing these graphs, users can easily monitor system performance, track changes, and ensure proper functioning of the overall setup during operation.

In addition, the graphical interface provides a clear timeline view, allowing users to analyze historical data and identify patterns or irregularities in system behavior. Sudden spikes or drops in the graph can indicate events such as unexpected motion or gate activation, which can be further investigated. The platform also supports data storage and retrieval, making it useful for future analysis and system improvements.

Fig 4.3 : Mobile Notification Output



This presents the output results of the proposed smart borehole safety system. The system successfully detects human or animal movement near the borehole and provides immediate safety responses such as LED indication, buzzer alerts, and automatic closure of the borehole cover using a servo motor. The outputs are clearly observed through the serial monitor, ThingSpeak graphs, and mobile notifications, even under varying environmental conditions. The inclusion of real-time data updates and alert messages enhances the reliability and usability of the system.

4.3 Discussion

The experimental results demonstrate that the proposed IoT-based smart borehole safety system effectively performs real-time monitoring, detection, and automated protection. The integration of multiple PIR sensors with the ESP32 microcontroller ensures accurate detection of human or animal movement, reducing false alarms through multi-sensor validation. The system exhibits fast response time by immediately activating safety mechanisms such as LED indicators, buzzer alerts, and automatic closure of the borehole cover using a servo motor. This rapid response is crucial in preventing accidents and enhancing safety in hazardous environments. Additionally, the system maintains continuous monitoring, ensuring reliable operation under different conditions.

The communication module further enhances the system by enabling real-time data transmission to the ThingSpeak platform and sending instant mobile notifications via Telegram. This allows users to remotely monitor system status and respond quickly in case of danger. Although the system performs efficiently, certain limitations such as dependency on Wi-Fi connectivity and sensitivity of sensors to environmental factors are observed. Despite these challenges, the overall system proves to be reliable, efficient, and suitable for practical implementation. The combination of sensing, processing, actuation, and communication makes the system a robust and scalable solution for borehole safety and accident prevention.

CHAPTER 5

CONCLUSION AND FUTURE WORK

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The proposed IoT-based smart borehole safety system successfully addresses the critical issue of accidents caused by open and unattended boreholes. The system integrates PIR sensors, an ESP32 microcontroller, and an automated actuation mechanism to provide real-time monitoring and immediate safety response. It effectively detects human or animal movement near the borehole and activates alerts such as buzzer and LED indications, along with automatically closing the borehole cover using a servo motor. The integration of IoT technology enables remote monitoring through the ThingSpeak platform and instant notifications via mobile alerts, ensuring timely awareness and action.

The experimental results confirm that the system provides high detection accuracy, fast response time, and reliable performance under different conditions. The multi-sensor approach reduces false detections, while the automated protection mechanism minimizes the need for human intervention. Overall, the system proves to be an efficient, cost-effective, and scalable solution for enhancing safety in hazardous environments, particularly in rural and construction areas.

5.2 Future Work

In future, the system can be enhanced by adding AI-based risk prediction, camera surveillance, solar power support, and GSM emergency alerts. Mobile app integration, cloud data analytics, and stronger weatherproof hardware can further improve performance and scalability for large-scale deployment.

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MAPPING – PO, PSO, AND SDG

PROGRAM OUTCOME (PO) MAPPING

The table below maps the outcomes of this project to the twelve Program Outcomes defined by the National Board of Accreditation. The mapping uses a correlation scale of 1 (low), 2 (medium), or 3 (high), indicating how strongly the project activities contributed to each outcome.

Program Outcome (PO) Mapping

PO NO.	Program Outcome	Relevance to Project	Level
PO1	Engineering Knowledge	Application of IoT concepts, embedded systems, sensors (PIR, ultrasonic), and ESP32 microcontroller to design a smart borehole safety system.	H
PO2	Problem Analysis	Identification of risks caused by open boreholes and analysis of limitations in traditional manual safety methods.	H
PO3	Design / Development of Solutions	Design and implementation of an automated system with sensing, processing, actuation, and communication modules for real-time safety.	H
PO4	Conduct Investigations of Complex Problems	Testing system performance under different conditions such as motion detection accuracy, response time, and alert generation.	H
PO5	Modern Tool Usage	Use of Arduino IDE, ESP32 programming, ThingSpeak cloud	H

		platform, and IoT communication technologies.	
PO6	The Engineer and Society	The system enhances public safety by preventing accidents in rural and construction areas, protecting human and animal lives.	H
PO7	Environment and Sustainability	Helps in maintaining safe surroundings and preventing environmental hazards caused by unattended boreholes.	M
PO8	Ethics	Ensures responsible use of IoT data and system reliability without causing harm or false alerts.	M
PO9	Individual and Team Work	Development of different modules such as hardware setup, coding, IoT integration, and testing in a team environment.	H
PO10	Communication	Project results documented in reports and presented through demonstrations, charts (ThingSpeak), and alerts (mobile notifications).	H
PO11	Project Management	Planning and execution of hardware design, coding, testing, and integration phases within the project timeline.	M
PO12	Life-Long Learning	Learning IoT technologies, embedded systems, and real-time monitoring techniques beyond the academic syllabus.	H

Program Specific Outcome (PSO) Mapping

PSO NO.	Program Specific Outcome	Project Contribution	Level
PSO1	Apply IoT and embedded system techniques to solve real-world problems	Implemented an IoT-based smart borehole safety system using ESP32, PIR sensors, ultrasonic sensors, and automated control mechanisms to prevent real-world accidents.	H
PSO2	Develop real-time monitoring and data-driven applications using modern tools	Developed a complete system using Arduino IDE, ESP32 programming, ThingSpeak cloud platform, and Telegram notifications for real-time monitoring and alerts.	H

SUSTAINABLE DEVELOPMENT GOAL (SDG) MAPPING

The following section outlines how this project aligns with the United Nations Sustainable Development Goals (SDGs), highlighting its relevance to each goal and explaining the specific contributions it makes toward achieving them.

Sustainable Development Goal (SDG) Mapping

SDG NO.	SDG Title	Project Relevance	Level
SDG 3	Good Health and Well-being	Prevents accidents and injuries caused by open boreholes by providing early detection, alerts, and automated protection mechanisms.	H
SDG 9	Industry, Innovation and Infrastructure	Introduces an innovative IoT-based safety system using sensors, ESP32, and automation to improve infrastructure safety and monitoring.	H
SDG 11	Sustainable Cities and Communities	Enhances safety in rural and urban areas by reducing risks associated with unattended boreholes and improving public safety systems.	H
SDG 12	Responsible Consumption and Production	Promotes efficient use of resources by implementing an automated system that reduces the need for manual monitoring and maintenance.	M
SDG 15	Life on Land	Protects both humans and animals from falling into boreholes, contributing to the safety of terrestrial life.	H

CONFERENCE DETAILS

CONFERENCE CERTIFICATE

Dr. P. Praveen Kumar, V. Geethamani, M.S Gopika, R.Rekha Sree from Kamaraj College of Engineering and Technology, Virudhunagar participated and presented a paper entitled “Smart Borehole Safety System” at the 10th International Conference on Engineering Technology and Science (ICETS’26) organized by Muthayammal College of Engineering, Rasipuram, Namakkal District, Tamil Nadu on March 28, 2026.



MUTHAYAMMAL COLLEGE OF ENGINEERING

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CERTIFICATE OF PARTICIPATION

This is to certify that

Praveen Kumar PremKamal

Kamaraj College of Engineering and Technology, Virudhunagar

has presented a paper titled

SMART BOREHOLE SAFETY SYSTEM

in

10th International Conference on Engineering Technology and Science (ICETS' 26)

organized by Muthayammal College of Engineering, Rasipuram,

Namakkal Dt., Tamil Nadu on 28th March 2026.



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